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Metallaphotoredox-catalyzed C–H activation: regio-selective annulation of allenes with benzamide†‡

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We have developed an efficient annulation of benzamides with allenes using cobalt and photoredox dual catalysis under an oxygen atmosphere. This reaction provides a mild and environmentally friendly method for the construction of isoquinolinone scaffolds in good to excellent yields, demonstrating broad substrate scopes, high regioselectivity, and good functional group compatibility. Notably, the transformation features an alternative strategy for the regeneration of a cobalt catalyst with the aid of Eosin Y. Preliminary mechanistic studies reveal that a radical-mediated cascade annulation is involved in this reaction.

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Introduction

Transition-metal-catalyzed direct C–H functionalizations have emerged as efficient, powerful, and straightforward approaches for the construction of carbon–carbon and carbon–heteroatom bonds over the past few decades.[1] Recently, low-cost and earth-abundant first-row transition metals came to prominence as an appealing alternative to noble metals.[2] Among the first-row transition metals, the unique properties of cobalt in varied oxidation states have witnessed the powerful ability to manipulate various chemical bonds as a result of the chelation-assisted direct metalation.[3] Among such elegant transformations, a number of C–H activation and oxidative annulation reactions with various unsaturated coupling partners, such as alkynes and alkenes, have garnered considerable attention for building molecule diversity *via* step- and atom-economical strategies.[4]

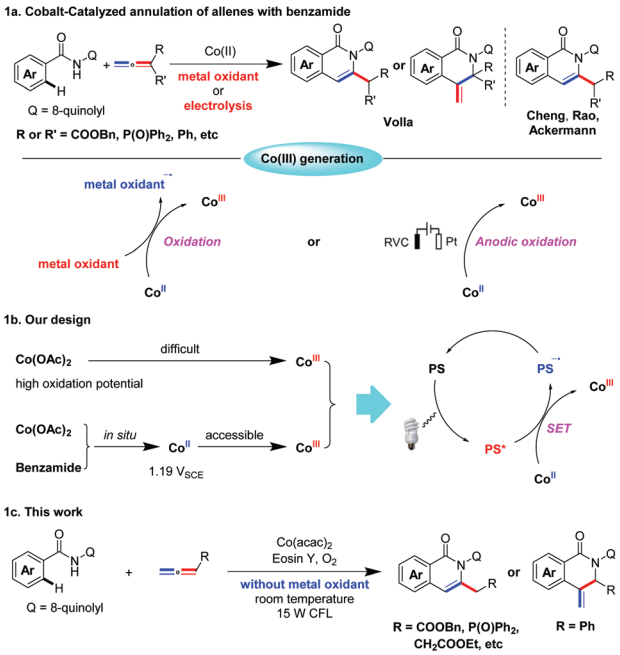
Allenes, bearing two orthogonal cumulative carbon–carbon double bonds, have been applied to a wide variety of annulations.[5] Various noble second- and third-row transition metals such as Pd,[6] Ir,[7] Rh,[8] and Ru,[9] have been employed for the preparation of cyclic scaffolds from simple, nonfunctionalized acyclic starting materials. Moreover, the first-row transition metals, such as Mn,[10] Fe,[11] Ni,[12] and Co,[13,14] have been recognized as cheap, environmentally friendly and effective catalysts for C–H functionalization with allenes. Particularly, cobalt-catalyzed regioselective intermolecular cyclization reactions of arylamides or alkenylamides with allenes were independently reported by the group of Volla,[14a] Rao,[14c] Cheng,[14e] Zhai,[14g] and Ackermann,[15] which provided rapid synthetic routes for diverse isoquinoline scaffolds. Notably, different reactivity patterns for the annulation could be observed in Volla's work through tuning electronic properties of the allenes, which led to efficient and rapid construction of dihydroisoquinolin-1(2*H*)-ones, isoquinolin-1(2*H*)-ones, *etc*. (Scheme 1a). Generally, stoichiometric amounts of metal oxidants (Ag_2CO_3, $Mn(OAc)_2$, $Mn(OAc)_3$, *etc*.), which could re-oxidize a low valent cobalt and complete the reaction cycle, were commonly employed in these reported cobalt-catalyzed annulation reactions of arylamides with allenes (Scheme 1a).[13,14] Recently, the Ackermann group disclosed an electrochemical C–H activation with allenes which enabled the high valent cobalt regeneration by anodic oxidation (Scheme 1a).[15]

In recent years, significant progress has been achieved based on visible light-mediated photoredox catalysis,[16] which emerged as a sustainable and versatile platform for the development of practical synthetic methodologies. Among these, the merge of transition metal catalysis and photocatalysis,

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Scheme 1 Catalytic approaches for C–H activation and annulation reactions with allenes.

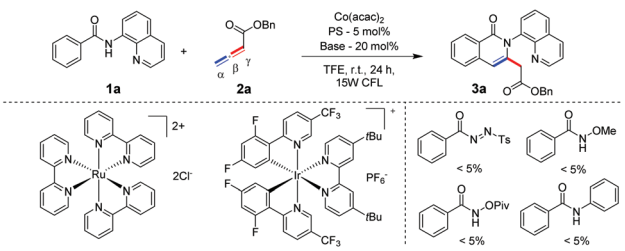
termed metallaphotoredox catalysis, affords a distinct reaction environment, with both oxidation and reduction of organic and organometallic species simultaneously possible.[17] On the basis of the huge advantage of metallaphotoredox catalysis in organic synthesis, we imagined that the development of cobalt-catalyzed C–H annulation of allenes with benzamide could be an important complement to the reported strategies.

Previous cyclic voltammetry study revealed $Co(OAc)_2$ to have a high oxidation potential, which meant the cobalt(III) could not be easily generated from Co(II) with common photocatalysts. However, the *in situ* generated cobalt catalyst between $Co(OAc)_2$ and benzamide substrate showed a lower oxidation potential at 1.19 V *vs.* SCE,[18] indicating an accessible photoredox mediated cobalt(II/III) single-electron oxidation (Scheme 1b).[19] Thus, we sought to regenerate the cobalt catalyst through a single electron transfer (SET) process assisted by a photocatalyst under the irradiation of light, which might avoid the use of metal oxidants (Scheme 1b). Herein, we report the application of a cobalt and photoredox dual catalysis strategy to the annulation of benzamides with allenes, which provides an economic and atom-efficient approach for the synthesis of isoquinolinones (Scheme 1c).

Results and discussion

As illustrated in Table 1, benzamides **1a** and benzyl buta-2,3-dienoate (**2a**) were exploited as model substrates with 10 mol% $Co(acac)_2$, 0.2 equiv. of NaOPiv and 5.0 mol% photoredox catalyst under visible light irradiation (15 W CFL) for 24 h at room temperature under an oxygen atmosphere. The model reaction was preferentially performed using 2,2,2-trifluoroethanol (TFE) as a solvent, mainly because of its ability to pair with hydrogen-bond acceptor groups and interfere positively with the catalytic cycle.[20] To our delight, the desired product **3a** could be obtained when $[Ru(bpy)_3]^{2+}Cl_2$ was used as a photoredox catalyst (entry 1). It was found that $[Ir(dF(CF_3)ppy)_2(dtbbpy)]PF_6$ did not effectively promote this cyclization (entry 2).

Table 1 Optimization the reaction conditions



Entry	[Co] (mol%)	PS (mol%)	Atmosphere	Base (mol%)	Time (h)	Yield[a] (%)
1	$Co(acac)_2$ (10)	[Ru] (5)	O_2	NaOPiv (20)	24	32%
2	$Co(acac)_2$ (10)	[Ir] (5)	O_2	NaOPiv (20)	24	<5%[b]
3	$Co(acac)_2$ (10)	Eosin Y (5)	O_2	NaOPiv (20)	24	39%
4	$Co(acac)_2$ (10)	Eosin Y (5)	O_2	KOAc (20)	24	41%
5	$Co(acac)_2$ (10)	Eosin Y (5)	O_2	KOTf (20)	24	55%
6	**$Co(acac)_2$ (20)**	**Eosin Y (5)**	**O_2**	**KOTf (20)**	**24**	**87%**
7	$Co(acac)_2$ (20)	Eosin Y (5)	Air	KOTf (20)	24	53%
8	$Co(acac)_2$ (20)	Eosin Y (5)	N_2	KOTf (20)	24	<5%[b]
9	—	Eosin Y (5)	O_2	KOTf (20)	24	<5%[b]
10	$Co(acac)_2$ (20)	—	O_2	KOTf (20)	24	16%
11	$Co(acac)_2$ (20)	Eosin Y (5)	O_2	KOTf (20)	24	14%[c]

All reactions were carried out under oxygen atmosphere unless otherwise stated, using **1a** (0.2 mmol), **2a** (0.4 mmol), cobalt catalyst (10–20 mol%), additive (20 mol%), 2,2,2-trifluoroethanol (2.0 mL) at room temperature under visible light irradiation (15 W CFL) for 24 hours. [a] Isolated yield. [b] Yield determined by ^1H NMR analysis using 4-nitrobenzaldehyde as the internal standard. [c] Without light.

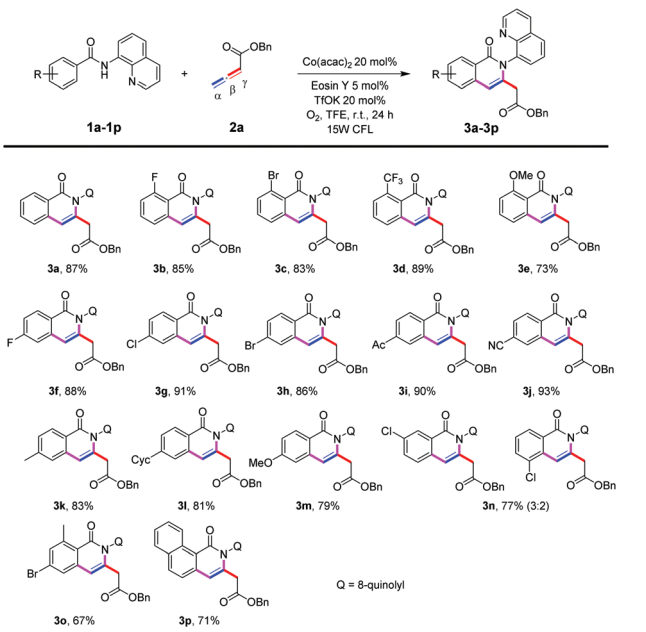
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Compared with $[Ru(bpy)_3]^{2+}Cl_2$, Eosin Y slightly increased the annulation yields under the similar reaction conditions (entry 3). Further experiment revealed that the use of KOTf as a base could accelerate the reaction and effectively improve the yield of **3a** (entries 4 and 5). Interestingly, a slight increase in the amount of cobalt had a significant effect in promoting the reaction, and the excellent yield was obtained when the annulation reaction was performed with 20 mol% $Co(acac)_2$ under the similar reaction conditions (entry 6). In addition, control experiments indicated that in the absence of photocatalyst, $Co(acac)_2$ or oxygen, the reaction was strongly inhibited (entries 7–10). It is worthy to note that light played a crucial role for the annulation reaction, which stopped at 14% yield when the reaction was performed in dark (entry 11). When benzamides with a weaker directing group were employed instead of **1a**, the desired product was not obtained under standard conditions. In these cases, the starting materials were fully recovered or decomposed.

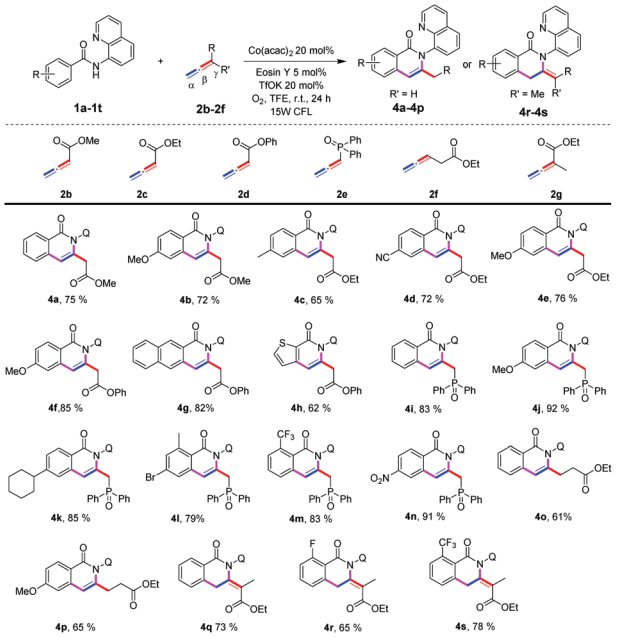
With the optimized conditions in hand, the annulation reaction scope was investigated using various substituted benzamides and allenes, and the results were presented in Scheme 2. We found that a number of arylamides bearing different substitutions at the *ortho*-, *meta*-, and *para*-positions were compatible with the optimized conditions (Scheme 2, **3a–3p**). For *para*-substituents in the carboxamides, such as halogen (**3f–3h**), acetyl (**3i**), cyano (**3j**), or electron-donating methyl (**3k**), cyclohexyl (**3l**), methoxyl (**3m**) groups, the desired isoquinolinones were produced in excellent yields. It was found that the reaction of *meta*-substituted benzamide furnished two expected regioisomers in good yields (**3n**). On the other hand, the substrates incorporated with electron-withdrawing *ortho*-substituents, such as halogen (**3b–3c**) and trifluoromethyl (**3d**) groups, were efficiently transformed into the corresponding products in excellent yields. With an *ortho*-methoxy substrate (**3e**), the cyclization process took a short reaction time, but a slightly reduced product yield was observed. In the case of disubstituted benzamide (**3o**), the reaction also gave the desired product in good yield. Additionally, the naphthoamide (**3p**) was smoothly converted into the three ring-fused heterocycle in good yield.

Next, the scope of allenes was investigated extensively. As summarized in Scheme 3, various allenes reacted smoothly with benzamides under the optimized conditions to produce the corresponding isoquinolinones in good to excellent yields. Notably, it was found that the cyclization reaction featured excellent chemo- and regioselectivity, and solely occurred at the terminal position of allenes. Various electronically diverged allenes (**2b–2f**) were efficiently converted into the desired products (**4a–4p**). A heteroaromatic substrate also showed good reactivity, and thieno[3,2-*c*]pyridin-4(5*H*)-one derivative (**4h**) was efficiently obtained in good yield. Gratifyingly, excellent yields (**4i–4n**) were obtained when diphenyl(propa-1,2-dien-1-yl)phosphine oxide (**2e**) was used. In the case of ethyl penta-3,4-dienoate (**2f**), the desired isoquinolinones (**4o** and **4p**) were afforded as single regioisomers with moderate yields. Interestingly, when a terminal disubstituted allene (**2g**) was used as a coupling partner, the reaction only afforded the corresponding *exo*-cyclic isoquinolinones (**4q–4s**) in moderate to good yields. These results provide useful information for the mechanistic perspective on the migratory allene insertion/isomerization manifold.

Generally, the formation of either alkenyl-metal or metal-π-allyl intermediates is dependent on the nature of allenes and metals in the transition-metal-catalyzed C–H



Scheme 2 Scope of benzamides with benzyl buta-2,3-dienoate.



Scheme 3 Scope of isoquinolinones from different allenes.

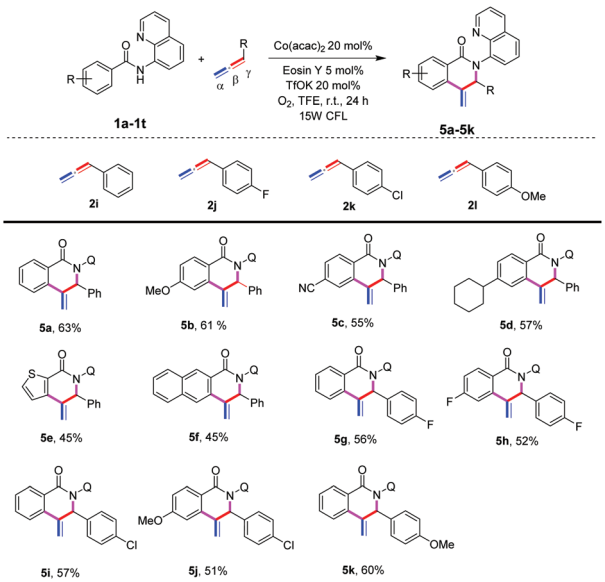
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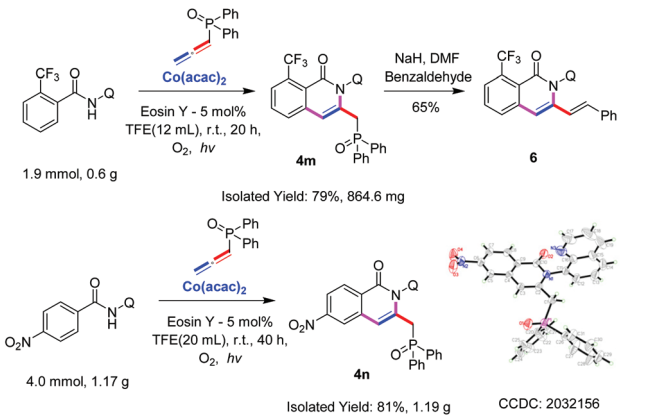
functionalization.[5,14,15] Volla and co-workers first disclosed that the regioselectivity of this annulation is dominated by the coordinative insertion of the double bond of an allene in cobalt-catalyzed reactions and that the electronical properties of allenes have a great influence on the reactivity of annulation.[14a] To our delight, the same results were obtained in our system. The carbocobaltation of phenylallene occurred at the more electron-rich double bond *via* formation of a stable Co-σ-allyl complex.[14a] Various isoquinolinones with an *exo*-double bond were obtained by coupling benzamides with propa-1,2-dien-1-ylbenzene (**2i**) under the standard conditions (Scheme 4). A number of ary amides (**5a–5d**) and heterocyclic derivatives, such as thiophene (**5e**), showed good compatibility with the reaction condition. Moreover, we investigated substituted phenylallenes such as **2j**, **2k** and **2l** with different benzamide, and the reaction provided the desired products in moderate yields.

A gram-scale reaction was conducted to assess the efficiency of this protocol (Scheme 5). Two isoquinolinones were furnished by employing 20 mol% $Co(acac)_2$ and 2.0 equiv. of preformed allene **2e** in 79% and 87% yield, respectively (for details, please see ESI‡). The structure and regioselectivity of the nitro substituted isoquinolinone (**4n**) were confirmed unambiguously by the X-ray crystallography. As shown in the post-annulation Wittig reaction, the cyclization product (**4m**) was efficiently transformed into olefin **6** with good yield.

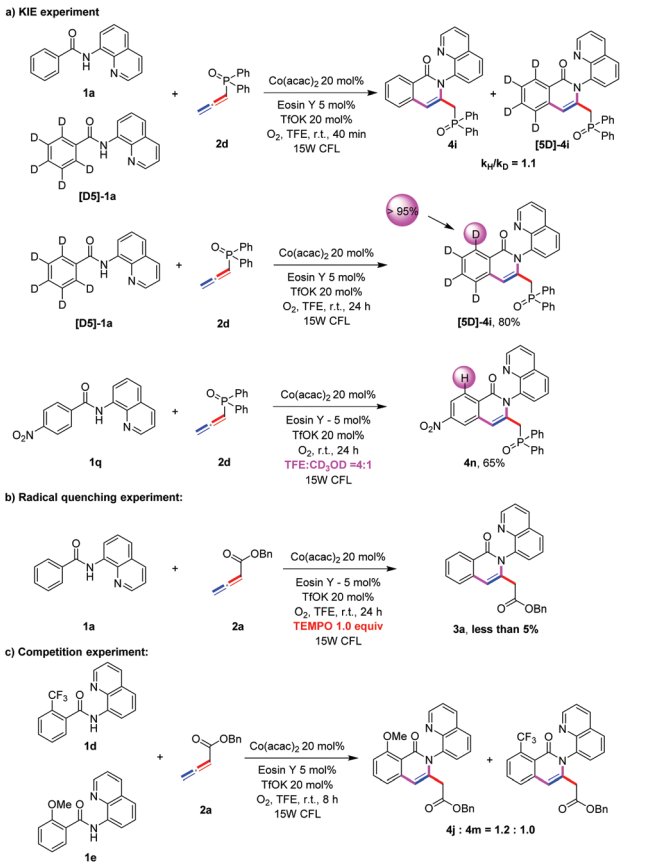
To gain further mechanistic insights, a series of control experiments were carried out. A mixture of benzamides [D5]-**1a**/**1a** (1 : 1) was used to react with allenylphsphine oxide **2e** under optimized conditions, and the KIE experiments gave a k_H/k_D value of 1.1. This result indicated that C–H bond activation might not be involved in the rate-determining step (Scheme 6a). Furthermore, no H/D exchange was observed



Scheme 4 Synthesis of isoquinolinones with *exo*-double bonds.



Scheme 5 Gram scale synthesis of dihydroisoquinolin-1(2*H*)-ones and application.



Scheme 6 Preliminary mechanistic investigation.

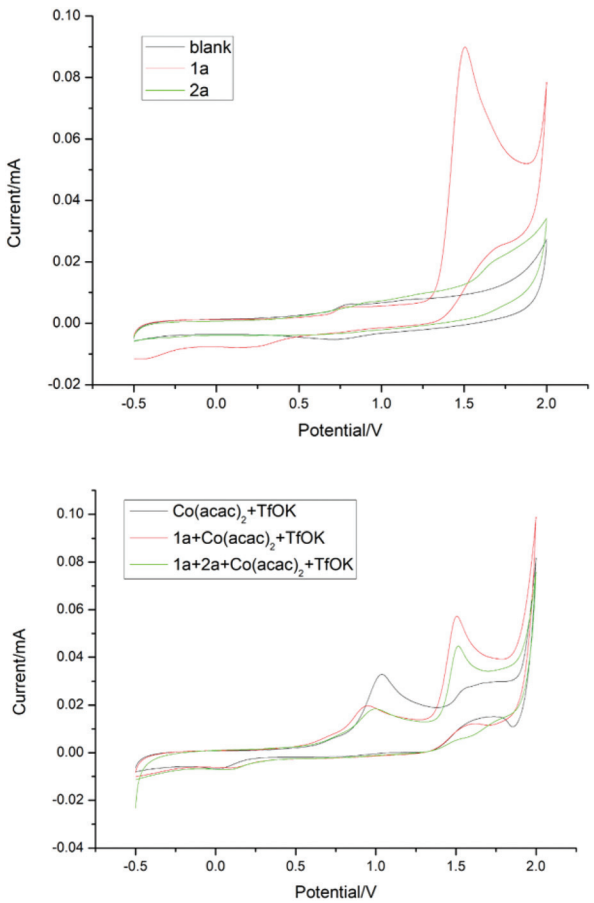
when [D5]-**1a** was treated with phosphinoylallene under standard conditions. Similarly, no deuterium incorporation in the product **4n** was observed when **1c** was treated with isotopically labelled CD_3OD as a cosolvent. These results suggested that the C–H cobaltation step should be irreversible (Scheme 6a). A control reaction performed with stoichiometric TEMPO (2,2,6,6-tetramethyl-1-piperidinyloxy, a common radical sca-

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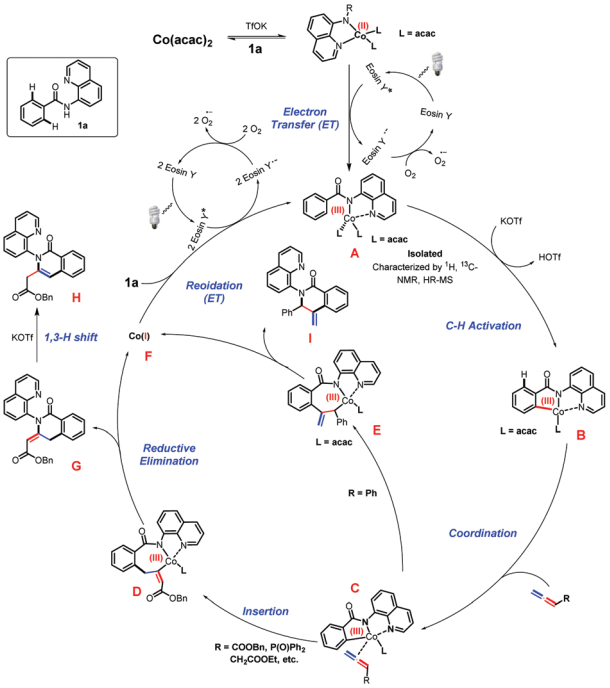
venger) formed no desired product under the standard condition (Scheme 6b). This gave a clue that a highly reactive radical species might be involved in the catalytic cycle. Additionally, the cyclization reaction conducted under an oxygen atmosphere in the absence of Eosin Y did not generate the desired product. It excluded the possibility of oxygen acting as a single oxidant that regenerated the cobalt catalyst. On the other hand, the result of the intermolecular competition experiment showed that the electron-rich amide **1e** was favorable over **1d** only slightly, suggesting little electronic influence in the C–H metalation (Scheme 6c).

Furthermore, detailed cyclic voltammetry studies were carried out to gain a better understanding of this annulation reaction. As shown in Scheme 7, the oxidation potential of substrate **1a** was 1.52 V *vs.* SCE, and **2a** had no relevant oxidation potential. The *in situ* generated cobalt catalyst[18] revealed a lower oxidation potential at 0.94 V *vs.* SCE, which might be oxidized by the photoexcited Eosin Y* (0.83 V *vs.* SCE).[21] These results demonstrated that a Co(II/III) single electron oxidation might occur in the initial step of this reaction.

Based on the mechanistic experiments described above, a plausible mechanism for this reaction was proposed in



Scheme 7 Cyclic voltammetry investigation. General conditions: Acetonitrile, 0.1 M *n*-Bu_4NPF_6, 5 mM TfOK, 5 mM $Co(acac)_2$, 5 mM substrates **1a** and **2a**, and 100 mV s^{-1}.



Scheme 8 Proposed reaction mechanism.

Scheme 8. The first step begins with the ligand exchange to generate the cobalt(II)-benzamide from $Co(acac)_2$, followed by oxidation of Co(II) to Co(III) occurring simultaneously with reduction of Eosin Y*. Additional ligand exchange and coordination with **1a** furnishes Co(III)-intermediate **A**. Interestingly, this intermediate was observed by subjecting stoichiometric $Co(acac)_2$ to the standard conditions (for details, see ESI‡).[22] Subsequently, activation of the inert aryl C–H bond forms the key metallacycle Co(III) intermediate **B**, which gives **C** upon allene coordination. Regioselective insertion of the C–Co bond into a π-bond of the allene gives the seven membered cobaltacycle intermediate **D** or **E**. The selectivity of the migratory insertion depends on the steric and electronic characteristics of the substituted allenes. When allenes were substituted by an electronic-withdrawing group (R in **C**), the insertion took place in the less-substituted double bond, leading to intermediate **D**. When R was a phenyl group, the addition was apt to occur at a more substituted double bond, furnishing intermediate **E** with high selectivity. The C–N bond-forming reductive elimination gives the cyclic product (isoquinolidinone **I** or **G**) and the cobalt(I) intermediate **F**. Intermediate **F** formed as a result of the reductive elimination then undergoes coordination with **1a** and might be oxidized by the photoexcited Eosin Y* (0.83 V *vs.* SCE)[21] (the incipient cobalt catalyst from $Co(acac)_2$ and benzamide **1a** showed an oxidation potential of 0.94 V *vs.* SCE). Thus, a Co(III) species could be regenerated with simultaneous formation of a reductive Eosin Y radical anion (−1.06 V *vs.* SCE).[21] The Eosin Y radical anion might be oxidized to the ground state by O_2 to complete the photoredox cycle.[21] At last, the isoquinolidinone intermediate **G** formed

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from an allenoate undergoes isomerization to give the isoquinolinone product **H**.

Conclusions

In conclusion, we have developed an efficient protocol for the construction of isoquinoline scaffolds ones by merging cobalt (III)-catalyzed C–H functionalization with visible light mediated photoredox catalysis. Notably, no metal oxidants are required in this transformation. The reaction provides a mild and environmentally friendly route for the efficient synthesis of isoquinolinones and isoquinolidinones. A wide substrate scope, high level of chemoselectivity and regioselectivity, and broad functional group tolerance were observed. Preliminary mechanistic studies reveal that a radical-mediated cascade annulation is involved in this reaction. Further studies about synthetic applications and reaction mechanism are currently ongoing in our laboratory.

Author contributions

Tianlei Li, Jishun Li, Zihao Zhu, Yuyuan Chen, Xueyao Li, and Wenxuan Zhang conducted the experiments and prepared the ESI.‡ Tianlei Li, Jie Xia and Qingyun Yang characterized and analyzed the X-ray crystal data. Tianlei Li, Song Wu, Weidong Pan and Chao Zhang designed the experiments and wrote the paper. All authors have given approval to the final version of the manuscript.

Conflicts of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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